

Effects of Parameter Changes on Graphs of Quadratic Functions

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The **vertex form** of a quadratic function is

$$y = a(x - h)^2 + k.$$

Vertex

The **vertex** of the parabola is (h, k) .

$$y = a(x - h)^2 + k$$

Therefore,

$$\boxed{(h, k)}$$

Axis of Symmetry

The **axis of symmetry** is the vertical line

$$x = h.$$

It passes through the vertex (h, k) .

Parameter a

The parameter a determines the **opening direction** and the **width** of the parabola.

$$y = a(x - h)^2 + k$$

Effect of a

If $a > 0$, the parabola opens upward.

If $a < 0$, the parabola opens downward.

$$a > 0 \Rightarrow \text{upward}$$

$$a < 0 \Rightarrow \text{downward}$$

Width of the Parabola

If $|a| > 1$, the parabola becomes narrower.

If $0 < |a| < 1$, the parabola becomes wider.

$$|a| > 1 \Rightarrow \text{narrower}$$

$$0 < |a| < 1 \Rightarrow \text{wider}$$

Parameter h

The parameter h moves the graph horizontally.

Positive h shifts the graph to the right.

Negative h shifts the graph to the left.

Parameter k

The parameter k moves the graph vertically.

Positive k shifts the graph upward.

Negative k shifts the graph downward.

Examples

Example 1

Describe the effect of changing a from 1 to 3
in

$$y = a(x - 2)^2 - 1.$$

The parabola becomes **narrower**.

Example 2

Compare

$$y = (x - 1)^2 + 2$$

and

$$y = (x - 4)^2 + 2.$$

The graph shifts **3 units to the right**.

Example 3

State the vertex, opening direction, and axis of symmetry of

$$y = -2(x - 3)^2 + 5.$$

Vertex: $(3, 5)$

Opens downward, $x = 3$

Practice Exercises

Practice 1

Describe the effect of changing a from 1 to 3
in

$$y = a(x - 2)^2 - 1.$$

Practice 2

Compare the graphs of

$$y = (x - 1)^2 + 2$$

and

$$y = (x - 4)^2 + 2.$$

Describe the effect of changing h .

Practice 3

State the vertex, opening direction, and axis of symmetry of

$$y = -2(x - 3)^2 + 5.$$

Practice 4

Which parameter should be changed to make

$$y = (x + 4)^2 - 2$$

open downward without changing its vertex?

Write the new equation.

Practice 5

Determine the values of a , h , and k in

$$y = -\frac{3}{2}(x + 5)^2 - 4.$$

Then describe the graph.

Activities

Activity 1

Describe the effect of changing a from -1 to $-\frac{1}{3}$ in

$$y = a(x - 2)^2 - 4.$$

Activity 2

Compare the graphs of

$$y = (x - 5)^2 + 2$$

and

$$y = (x - 5)^2 - 3.$$

Describe the effect of changing k .

Activity 3

Which parameter should be changed to move the graph

$$y = (x + 2)^2 - 1$$

five units upward?

Write the new equation.

Activity 4

Explain how the graph changes when

$$y = \frac{1}{4}(x + 1)^2 - 3$$

is changed to

$$y = 5(x + 1)^2 - 3.$$

Activity 5

A parabola has vertex $(-3, 5)$, opens downward, and is wider than $y = x^2$.

Write one possible equation in vertex form.

Thank you!